Tennessee Valley Authority, Post Office Box 2000, Decatur, Alabama 35609-2000

April 24, 2017

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 2
Renewed Facility Operating License No. DPR-52
NRC Docket No. 50-260

Subject: **Licensee Event Report 50-260/2017-002-00**

The enclosed Licensee Event Report provides details of an inoperable Primary Containment Isolation Valve resulting in a condition prohibited by Technical Specifications. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(i)(B).

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact M. W. Oliver, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

S. M. Bono
Site Vice President

Enclosure: Licensee Event Report 50-260/2017-002-00 – Inoperable Primary Containment Isolation Valve Resulting in Condition Prohibited by Technical Specifications

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant

ENCLOSURE

Browns Ferry Nuclear Plant
Unit 2

Licensee Event Report 50-260/2017-002-00

Inoperable Primary Containment Isolation Valve Resulting in Condition Prohibited by Technical Specifications

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See Enclosed

NRC FORM 366 (04-2017) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020
LICENSEE EVENT REPORT (LER)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Browns Ferry Nuclear Plant, Unit 2	05000260	1 OF 7

4. TITLE
Inoperable Primary Containment Isolation Valve Resulting in Condition Prohibited by Technical Specifications

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME N/A	DOCKET NUMBER N/A
02	23	2017	2017	- 002	- 00	04	24	2017	FACILITY NAME N/A	DOCKET NUMBER N/A

9. OPERATING MODE	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
074	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Baruch Calkin, Licensing Engineer	256-614-6713

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	BN	ISV	W030	Y	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED		15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)*	☒ NO		N/A	N/A	N/A

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On February 23, 2017, radiography results for the Browns Ferry Nuclear Plant (BFN), Unit 2, Reactor Core Isolation Cooling (RCIC) system stop check valve, 2-HCV-071-0014 (71-14 valve), showed the valve to be in the fully open position. In this position, the valve was incapable of performing its design function as a Primary Containment Isolation Valve (PCIV). In order to isolate the affected primary containment penetration flowpath, the RCIC Steam Line Outboard Isolation Valve was closed and deactivated to secure flow through the adjacent PCIV check valve in the exhaust line in accordance with plant Technical Specifications (TS) Limiting Conditions for Operation (LCO) for Primary Containment Isolation.

It is assumed in the Past Operability Evaluation (POE) that the 71-14 valve had been inoperable from April 2015 to February 2017, in violation of TS LCO 3.6.1.3. During this time period, another check valve (upstream of the 71-14 valve) remained capable of isolating the affected penetration. The 71-14 valve was repaired during the BFN, Unit 2, Cycle 19 Refueling Outage which began on February 25, 2017.

The cause of this condition was determined to be inadequate minimum allowable internal clearances for the 71-14 valve. The 71-14 valve internals were machined to minimum allowable material as an interim action to mitigate risks. The corrective action to prevent recurrence is to replace the 71-14 valve with a new valve design that is not susceptible to the same internal binding potential.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 2	05000-260	2017	002	00

NARRATIVE

I. Plant Operating Conditions Before the Event

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Unit 2, was in Mode 1 at approximately 74 percent rated thermal power.

II. Description of Event

A. Event Summary:

On February 23, 2017, at approximately 2239 Central Standard Time (CST), radiography results for BFN, Unit 2, Reactor Core Isolation Cooling (RCIC) [BN] system stop check valve [ISV], 2-HCV-071-0014 (71-14 valve), was discovered to be in the fully open position, thereby not meeting the acceptance criteria of Surveillance Instruction 2-SI-3.2.3, Testing ASME OM Code and Augmented IST Check Valves (Internal Inspection). This valve is the RCIC turbine exhaust stop check valve and is one of two Primary Containment Isolation Valves (PCIVs) connecting the RCIC exhaust directly to primary containment. The acceptance criteria is for the valve to be fully closed to satisfy the containment isolation function. A Past Operability Evaluation (POE) determined that the 71-14 valve was inoperable from April 3, 2015, at 2356 Central Daylight Time (CDT), until Unit 2 was placed in Mode 4 on February 25, 2017, at 0506 CST.

BFN, Unit 2, Technical Specifications (TS) Limiting Condition for Operation (LCO) 3.6.1.3 requires each PCIV, except reactor building-to-suppression chamber vacuum breakers, to be Operable in Modes 1, 2, and 3, and when associated instrumentation is required to be Operable per LCO 3.3.6.1, “Primary Containment Isolation Instrumentation”. In response to this event, on February 23, 2017, at 2250 CST, Operations personnel entered TS 3.6.1.3 Required Action A.1 which required the RCIC system to be isolated within 4 hours to prevent flow through the RCIC Turbine Exhaust check valve, 2-CKV-071-0580 (71-580 valve). Based on the results from the POE, the 71-14 valve was inoperable longer than allowed by the TS. On February 24, 2017, at 0018 CST, Operations personnel isolated the RCIC system in accordance with plant TS by closing the RCIC Steam Line Outboard Isolation Valve 2-FCV-071-0003 (71-3 valve).

On February 25, 2017, the scheduled Unit 2 refueling outage commenced.

On March 24, 2017 at 1641 CDT, BFN completed repairs to the 71-14 valve and restored the RCIC penetration flowpath.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

The 71-14 valve was unable to perform its function as a PCIV because it was stuck in the open position.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 2	05000-260	2017	002	00

C. Dates and approximate times of occurrences

Dates & Approximate Times	Occurrence
April 3, 2015, at 2356 CDT	It is assumed in the POE that the 71-14 valve bound open immediately after the start of the RCIC Turbine during performance of 2-SR-3.5.3.4, RCIC System Rated Flow at Low RPV Pressure.
February 23, 2017, at 2239 CST	Radiography results for the 71-14 valve showed the valve to be in the fully open position.
February 23, 2017, at 2250 CST	Operations declared 71-14 valve inoperable and entered TS Action 3.6.1.3.A to secure flow through check valve 71-580.
February 24, 2017, at 0018 CST	Valve 71-3 was closed to secure flow through check valve 71-580 which also removed the RCIC system from service.
February 25, 2017, at 0001 CST	Unit 2 inserted a manual reactor scram and entered Mode 3 for the previously scheduled Unit 2 refueling outage.
February 25, 2017, at 0506 CST	Unit 2 entered Mode 4 where operability of the 71-14 valve was no longer required.
March 24, 2017, at 1641 CDT	Operations personnel restored the RCIC penetration flowpath following completion of repairs to the 71-14 valve.

D. Manufacturer and model number of each component that failed during the event

The failed component was the 71-14 valve. The component was model 5312WE, manufactured by the Walworth Company.

E. Other systems or secondary functions affected

The 71-14 valve is one of two PCIVs connecting the RCIC exhaust directly to primary containment. Containment was isolated by closing the 71-580 valve resulting in the RCIC system being unable to perform its design function.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 2	05000-260	2017	002	00

F. Method of discovery of each component or system failure or procedural error

On February 23, 2017, radiography results for the BFN, Unit 2, 71-14 valve showed the valve to be in the fully open position.

G. The failure mode, mechanism, and effect of each failed component

The disc of the 71-14 valve was stuck in the fully open position due to internal binding. Three factors aligned to cause the internal binding of the valve: limited internal clearance for moving parts, corrosion debris from the RCIC piping was deposited into the valve, and the valve bonnet was not centered on the body. This configuration resulted in the 71-14 valve being unable to perform its safety function as a PCIV.

H. Operator actions

Operations personnel declared the 71-14 valve inoperable. The Primary Containment flowpath was isolated by securing the 71-580 valve. The 71-3 valve was closed, resulting in the RCIC system becoming inoperable.

I. Automatically and manually initiated safety system responses

There were no safety system responses associated with this event.

III. Cause of the event

A. Cause of each component or system failure or personnel error

The root cause of this condition was inadequate minimum allowable clearances for the 71-14 valve.

There were two direct causes identified for the valve’s failure:

1. The Corrective Action to Prevent Recurrence developed for a similar 2013 valve failure, which was to submit a Preventive Maintenance Change Request to open, inspect, clean, and replace valve disc, dashpot and stem as necessary for Units 1, 2, and 3 71-14 valves, did not prevent recurrence of future similar events.
2. The 71-14 valve was mechanically bound, preventing the valve disc from returning to the closed position.

B. Cause(s) and circumstances for each human performance related root cause

There were no human performance root causes.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 2	05000-260	2017	002	00

IV. Analysis of the event

The Tennessee Valley Authority (TVA) is submitting this report in accordance with Title 10 of the Code of Federal Regulations (10 CFR) 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS.

The 71-14 valve is one of two PCIVs connecting the RCIC exhaust directly to primary containment through penetration X-212. This 71-14 valve has a safety-related function in the closed position and a quality-related, important to safety function, in the open position. According to TS Bases 3.6.1.3 for PCIVs, PCIVs form a part of the primary containment boundary. The PCIV safety function is related to minimizing the loss of reactor coolant inventory and establishing the primary containment boundary during a Design Basis Accident. PCIV operability supports leak tightness of primary containment.

The 71-14 valve is normally closed to provide its safety-related function as containment isolation. The valve opens from its normally closed position to allow a flow path from the RCIC Turbine Exhaust to the Pressure Suppression Chamber (PSC). The 71-14 valve closes when the RCIC system is shut down due to the weight of the valve disc. The exhaust header is designed to contain a check valve and a stop check valve to prevent backflow of Suppression Pool water into the RCIC exhaust line.

TS LCO 3.6.1.3 for PCIVs requires each PCIV, except reactor building-to-suppression chamber vacuum breakers, to be Operable in Modes 1, 2, and 3. However, on February 23, 2017, the 71-14 valve was found failed open during surveillance testing. At the time of discovery, Unit 2 was in Mode 1. As a result of the 71-14 valve being declared inoperable, Operations personnel entered TS 3.6.1.3 required actions to isolate the RCIC system preventing flow through the 71-580 valve which is the upstream RCIC turbine exhaust line PCIV. The 71-14 valve was determined to have been inoperable longer than allowed by TS resulting in a condition prohibited by the plant's TS.

V. Assessment of Safety Consequences

The ability of the Primary Containment penetration X-212 to perform its design function was not adversely affected by the 71-14 valve being bound in the open position due to the 71-580 valve remaining capable of isolating the penetration.

Additionally, the ability of the 71-14 valve to fulfill its function of providing a flow path for RCIC Turbine Exhaust Steam was not adversely affected by it being bound in the open position. Since the 71-14 valve must be in the open position to operate RCIC, the RCIC System injection function was not affected when this valve was bound open; therefore, the RCIC System was capable of injecting water into the reactor vessel and would have met its mission time.

Therefore, the RCIC system was still capable of performing its required safety functions to protect the health and safety of the public.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

With the 71-14 valve being bound in the open position, this valve could not perform its design function as a PCIV. However, the 71-580 valve remained capable of isolating the penetration.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when Unit 2 reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

It is assumed in the POE that the 71-14 valve was inoperable from April 3, 2015, at 2356 CDT, until Unit 2 was placed in Mode 4 on February 25, 2017, at 0506 CST.

VI. Corrective Actions

Corrective Actions are being managed by TVA's corrective action program under Condition Report (CR) 1265552.

A. Immediate Corrective Actions

The 71-14 valve was repaired.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

The root cause of this event will be addressed by replacing the Unit 2 71-14 valve with a new valve design that is not susceptible to the same internal binding potential as the existing valve.

VII. Previous Similar Events at the Same Site

Previous similar events at BFN include failure of the Unit 2 71-14 valve. On March 2, 2013, radiography results for the Unit 2 71-14 valve displayed the valve bound in the fully open position. This event was reportable, and LER 50-260/2013-001-00, Inoperable Stop Check Valve Results in Condition Prohibited by Technical Specifications and Reactor Shutdown Required by Technical Specifications, was submitted to the Nuclear Regulatory Commission on May 1, 2013.

A search was performed of CRs relating to the 71-14 valve. CRs related to the condition reported in the above 2013 LER are CRs 689932, 696534, and 79167. The corrective actions from the Root Cause Analysis documented in CR 696534 did not prevent the recurrence of this event.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.